



Coffee Shop Sales Analysis

A data-driven analysis of coffee shop sales to understand customer behavior, optimize operations, and improve business performance

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Why Analyze Coffee Shop Sales?

A data-driven analysis of coffee shop sales to understand customer behavior, optimize operations, and improve business performance.

Rich Transactional Data

Coffee shops generate thousands of daily transactions — a goldmine for behavioral and operational insights.

Understanding Customers

Analytics reveals purchase patterns, peak demand windows, and category preferences across store locations.

Smarter Decisions

Data-driven insights directly improve staffing, inventory planning, and revenue optimization strategies.

Project Objectives

1

Sales & Revenue

Analyze total sales volume and revenue across the dataset.

2

Store & Product Comparison

Benchmark performance across store locations and product lines.

3

Peak Hour Identification

Pinpoint high-traffic time windows to inform staffing decisions.

4

Temporal Trends

Examine weekday vs. weekend behavior and month-over-month patterns.

Purpose & Scope

Purpose

This project extracts actionable business insights from raw point-of-sale data — translating transaction logs into clear, visual narratives that support operational and strategic decisions.

→ Data Preprocessing

Cleaning and structuring raw inputs for reliable analysis.

→ Exploratory Data Analysis

Surfacing trends, distributions, and anomalies in the data.

→ Feature Engineering

Creating derived variables to enrich downstream analysis.

→ Business Visualization

Communicating findings through clear, interpretable charts.

Dataset Overview



Source

Coffee Shop Sales dataset provided in Excel format, capturing granular transaction-level records across multiple store locations.

Key Fields

- **Transaction Date & Time** — precise timestamp per sale
- **Store Location** — identifies originating branch
- **Product Category & Type** — beverage, food, merchandise
- **Quantity & Unit Price** — inputs for revenue calculation

Tools & Technologies



Python

Core programming language for all data loading, transformation, and analysis tasks.



Pandas & NumPy

Data manipulation and numerical computation libraries for preprocessing and aggregation.



Matplotlib & Seaborn

Python visualization libraries used to generate all plots and business-metric charts.



Jupyter Notebook

Interactive development environment enabling reproducible, documented analysis.



Excel

Source format for the raw dataset; used for initial review and validation before Python ingestion.

Data Preprocessing



A systematic four-stage pipeline ensured the dataset was complete, consistent, and correctly typed before any analysis began.

Missing Value Treatment

Identified and handled null entries to maintain dataset integrity.

Date & Time Parsing

Converted raw string timestamps into structured datetime objects for temporal analysis.

Duplicate Removal

Detected and dropped duplicate records to prevent inflated metrics.

Revenue Column

Derived **Total Sales** = **Quantity** × **Unit Price** as the primary KPI.

Feature Engineering

New variables were derived from raw fields to unlock richer temporal and behavioral patterns in the data.

Time Decomposition

Extracted **Month**, **Day of Week**, and **Hour** from transaction timestamps to enable fine-grained trend analysis.



Weekday / Weekend Flag

Binary categorical variable distinguishing weekday and weekend transactions to study behavioral shifts.

Aggregated Metrics

Grouped data by store and product to generate revenue totals, transaction counts, and average order values.



Summary Statistics

Computed key descriptive metrics — mean, median, totals — used throughout the EDA and visualization phases.

Exploratory Data Analysis

The EDA phase investigated four core dimensions of the business to surface actionable patterns.



Store Performance

Revenue and transaction volume compared across all three locations.



Product Categories

Sales breakdown by beverage, food, and merchandise types.



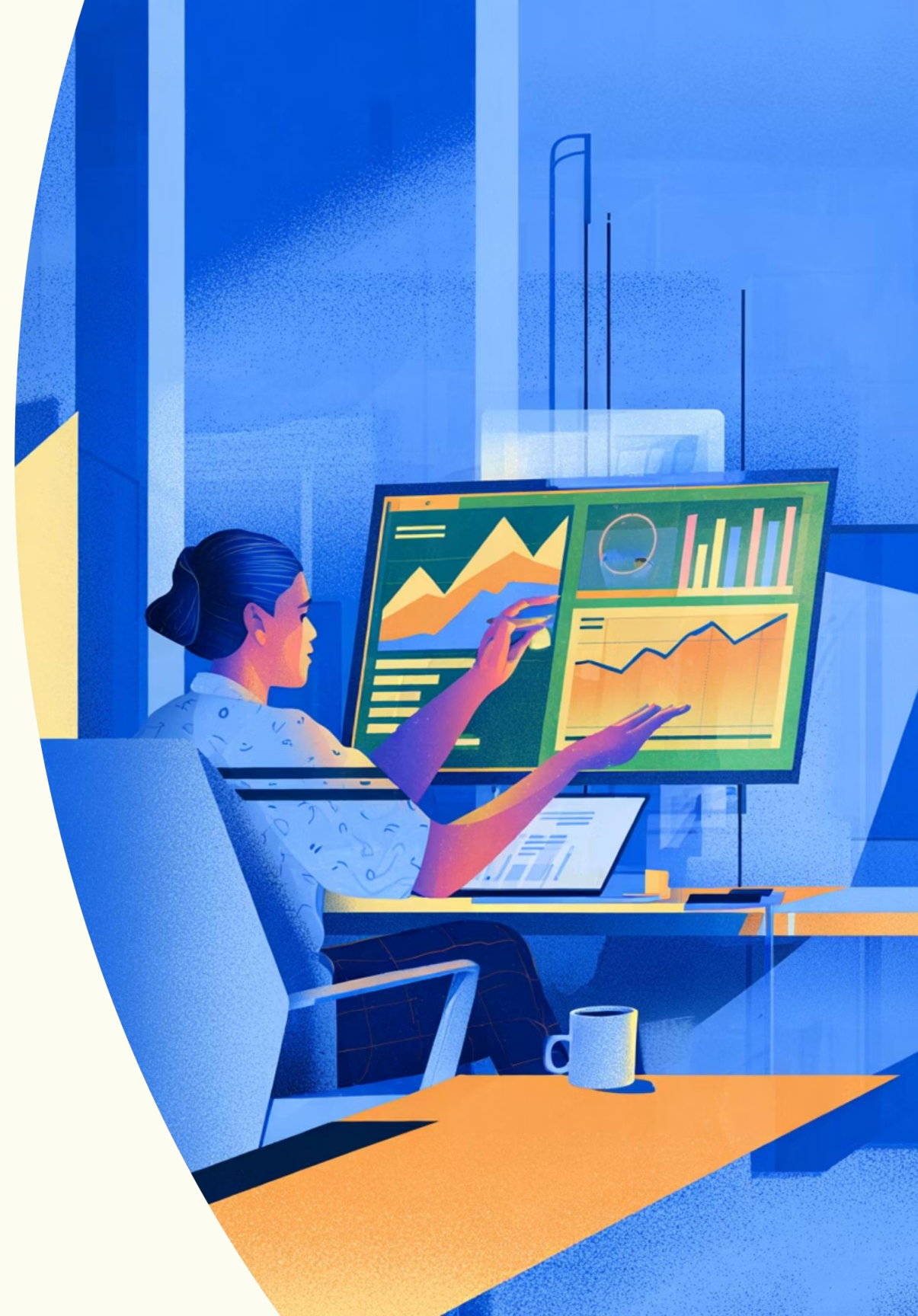
Time-Based Trends

Hourly and monthly patterns revealing demand peaks and troughs.

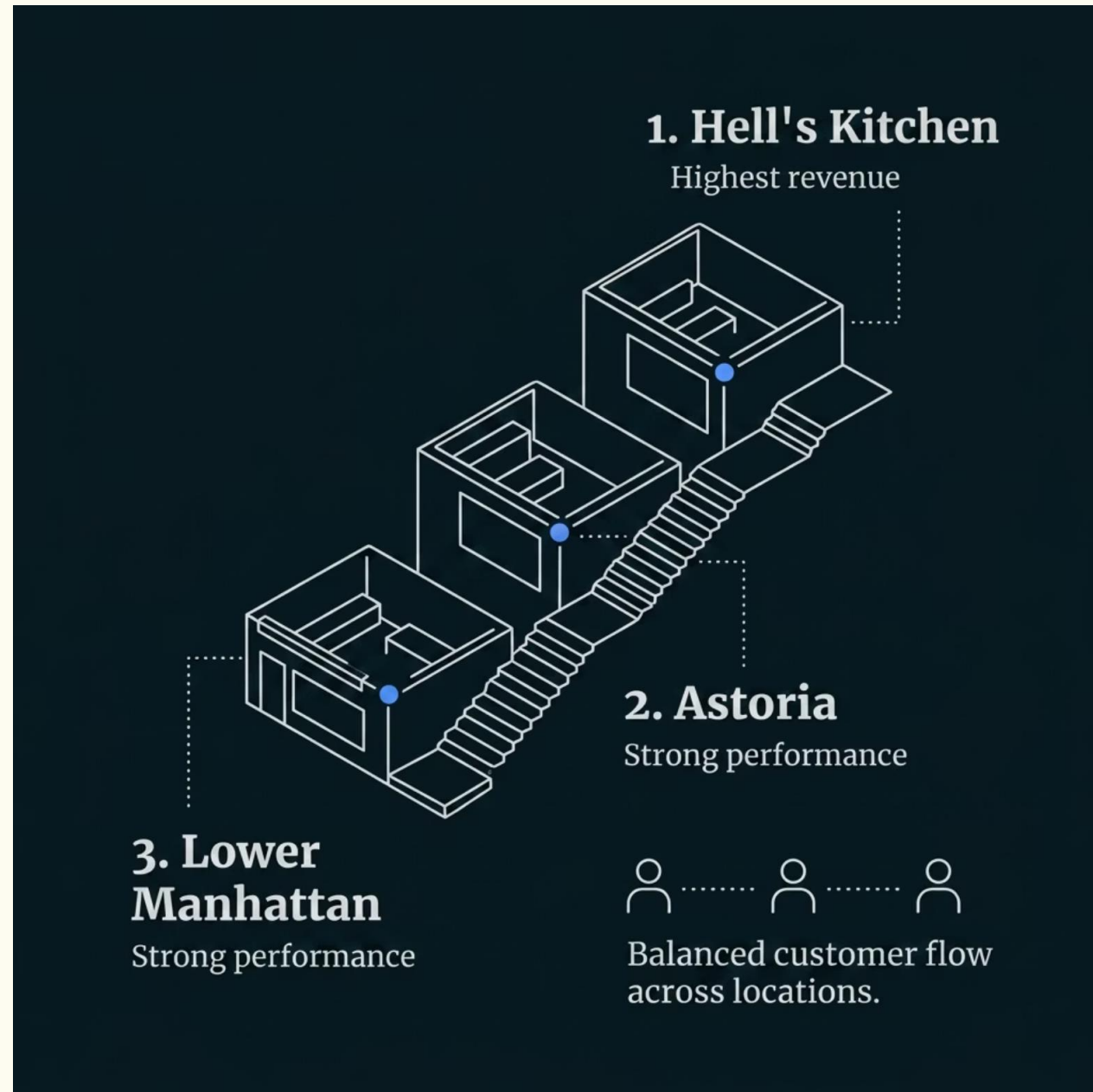


Customer Behavior

Weekday vs. weekend purchasing patterns and order frequency analysis.



Store Performance Highlights



Key Findings

Hell's Kitchen led all three locations in total revenue, driven by high foot traffic and strong average transaction values.

Astoria and **Lower Manhattan** both delivered competitive results, indicating a well-distributed customer base across the brand's footprint.

- ✓ All three stores maintained balanced customer flow — no single location showed signs of underperformance relative to its market context.



Product Performance

Analyzing sales data revealed clear leaders and key products driving revenue across all locations.

Coffee

Consistently the top-performing category, reflecting strong customer preference and diverse offerings.



Tea

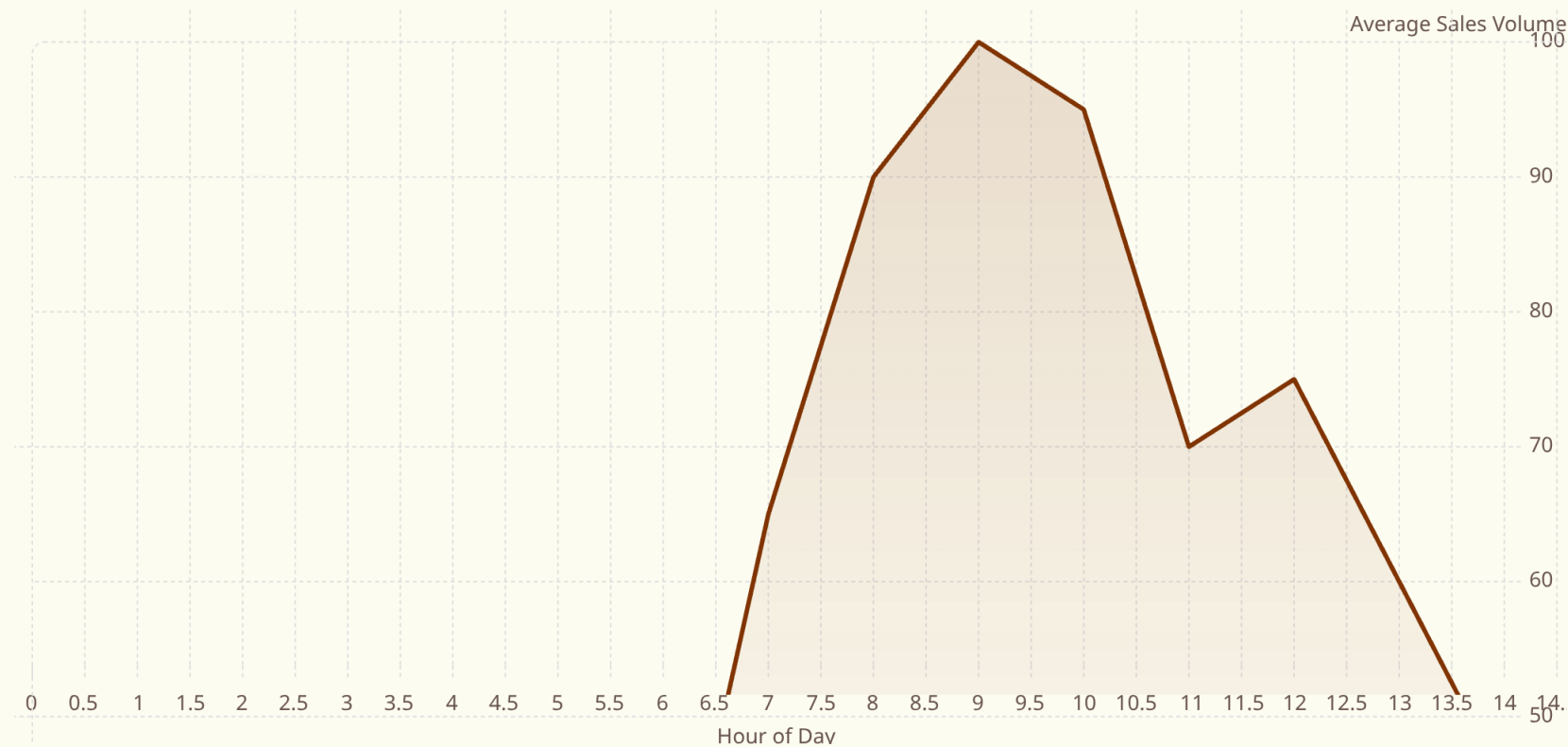
The second highest-selling category, with steady demand across various blends and seasonal specials.

Top-Selling Products

- **Sustainably Grown Organic:** Premium coffee blend, appealing to eco-conscious customers.
- **Dark Chocolate:** Popular choice for both hot and cold beverages, especially as an add-on.
- **Latte:** Classic espresso-based drink, maintaining strong sales volume throughout the day.

Time-Based Insights

Analyzing transaction timestamps reveals distinct patterns in customer activity throughout the day and across the week, highlighting key periods for operational focus.



The hourly sales volume data clearly indicates a significant peak between **8 AM and 10 AM**, with **10 AM** registering the highest sales throughout the day. Conversely, evening sales consistently show lower volumes, especially after 6 PM. Furthermore, total sales volume is notably higher on **weekdays compared to weekends**, suggesting a primary customer base of commuters and office workers.

Pricing & Sales Relationship

Main Finding:

Clear inverse relationship between price and quantity sold:

- **Low price** → high quantity
- **High price** → low quantity

Advanced Insight:

- Price-sensitive customers dominate
- Demand elasticity is high

Business Action:

- Use dynamic pricing strategy
- Keep entry-level products affordable
- Premium pricing only for specialty items

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Customer Behavior Analysis (Deep)

Key Finding:

- Weekdays >> Weekends (significantly higher weekday sales)
- Average order value ≈ \$4.69

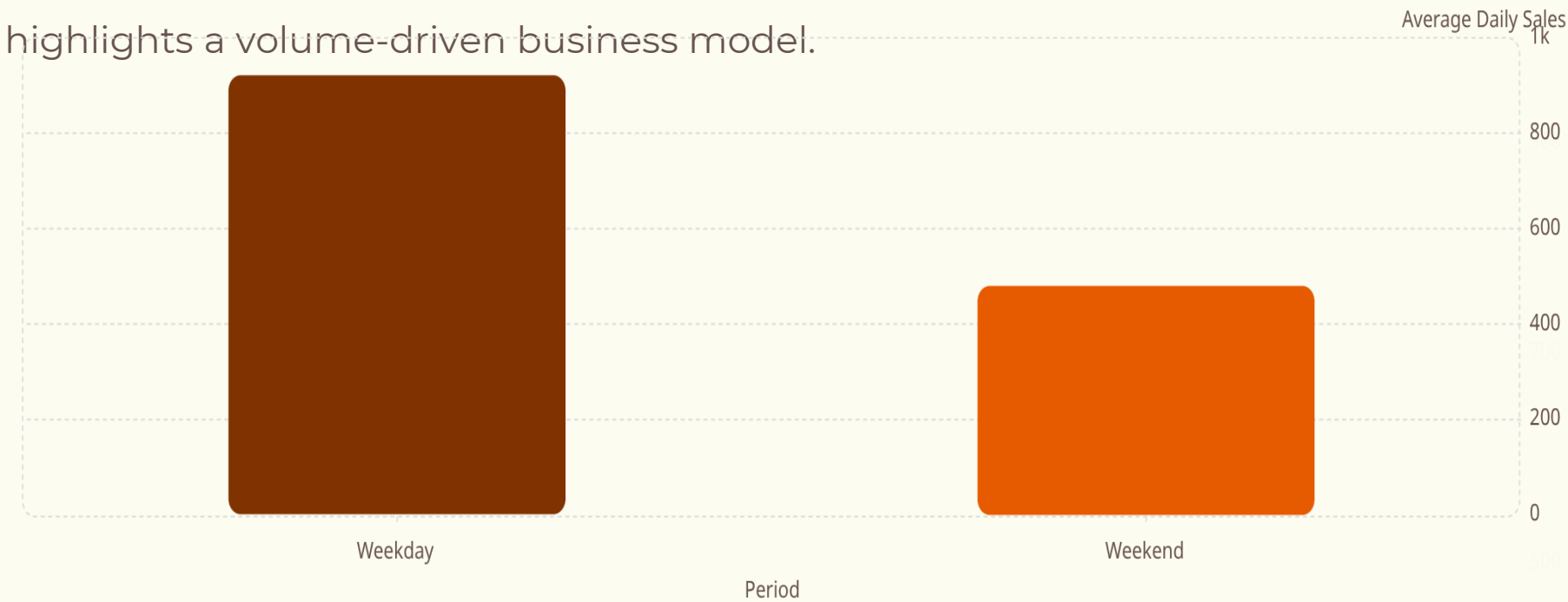
Advanced Insight:

- Business depends on high volume, low-value transactions
- Customers prefer quick, affordable purchases

Business Action:

- Increase order value using:
- Combo offers
- Upselling (add-ons like snacks)
-

Analyzing sales patterns reveals the clear dominance of weekday transactions over weekends, alongside an average order value that highlights a volume-driven business model.



Key Analysis Findings

Our deep dive into the sales data uncovered several critical insights about customer behavior and product performance, highlighting areas for strategic focus.

269,952+

Coffee Sales

Total units sold, consistently driving top-line revenue.

196,405+

Tea Sales

Robust demand for tea, positioning it as a strong secondary category.

≈\$4.69

Avg Order Value

Indicates customer spending patterns per transaction.

The analysis confirms this is a **high transaction volume business**, where efficient service is paramount. Furthermore, insights suggest that **pricing adjustments directly influence customer demand**, underscoring the importance of dynamic pricing strategies.

Final Insights

Our comprehensive analysis reveals key strategic recommendations to optimize sales and operational efficiency across all coffee shop locations.

★ Focus on Top Products

Prioritize inventory, marketing, and promotions for high-demand coffee and tea items to maximize revenue.

🏷️ Optimize Pricing Strategy

Implement dynamic pricing models and special offers to improve margins and attract customers during off-peak times.

📉 Improve Low-Performing Categories

Conduct targeted analysis on underperforming food or merchandise categories to identify issues and develop improvement plans.

🌙★ Increase Evening Sales

Explore evening promotions, happy hour specials, or community events to boost traffic after 6 PM.

👥 Align Staffing with Peak Hours

Adjust staffing schedules to match customer flow, ensuring efficient service during busy morning hours and cost-effectiveness during slower periods.



Key Recommendations

Our comprehensive analysis reveals key strategic recommendations to optimize sales and operational efficiency across all coffee shop locations.

 **Focus on High-Performing Products**
Prioritize inventory, marketing, and promotions for high-demand coffee and tea items to maximize revenue.

 **Optimize Pricing Strategy**
Implement dynamic pricing models and special offers to improve margins and attract customers during off-peak times.

 **Improve Low-Performing Categories**
Conduct targeted analysis on underperforming food or merchandise categories to identify issues and develop improvement plans.

 **Maximize Peak Hour Performance**
Adjust staffing and optimize service to maximize efficiency during busy morning hours (8 AM - 10 AM) to capitalize on highest sales volume.

 **Boost Evening Sales**
Explore evening promotions, happy hour specials, or community events to boost traffic after 6 PM.

 **Enhance Customer Experience**
Improve service efficiency and personalize customer interactions to foster loyalty and repeat business.

 **Increase Average Order Value**
Introduce bundle deals, upsell opportunities, and loyalty programs to encourage customers to spend more per transaction.

Data-Driven Decision Making

Utilize ongoing data analysis to continuously refine strategies and identify new growth opportunities.



THANK YOU